

Name: Justin M. Iadicola

Date: 5/11/2026

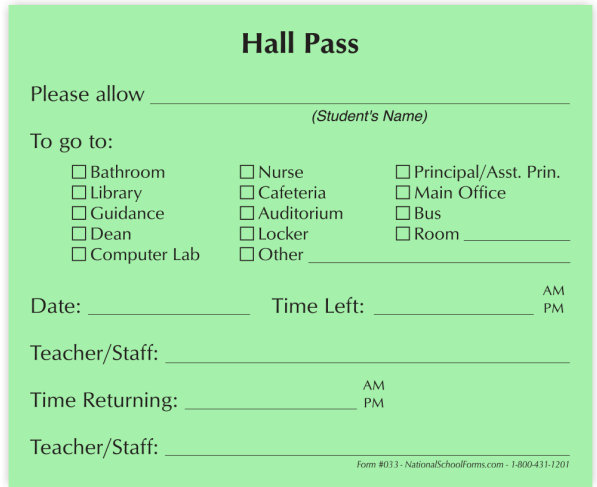
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MyDesign Portfolio

Element B

Element B: Documentation and Analysis of Current and Prior Design Attempts

Current or Prior Design Attempt #1 - Traditional Paper Hall Passes

Source	"Fast Facts: School Safety and Security Measures." National Center for Education Statistics
Image	 https://nationalschoolforms.com/school-forms/hall-passes/hall-pass-pad-green/
Details	Traditional paper hall passes are commonly used in middle and high schools to allow students to temporarily leave class for

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	destinations such as bathrooms, nurse offices, or counseling centers. Teachers typically write or hand students a paper pass that identifies the student and destination. This system is inexpensive and easy to implement because it requires no electronic hardware or software systems. However, according to the National Center for Education Statistics, schools continue increasing security and monitoring efforts due to concerns involving student supervision and accountability (National Center for Education Statistics, 2022).
<i>Analysis</i>	Paper hall pass systems are ineffective because they provide no real-time tracking or verification that a student actually reached the intended destination. Teachers must rely entirely on trust and memory, especially when multiple students leave class at the same time. Students can also misuse passes, remain outside class for excessive periods of time, or travel to unauthorized locations without detection. Although paper passes are simple and low cost, they fail to satisfy the project's design requirements for automated monitoring, time tracking, and student verification.



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Prior Design Attempt #2 - SmartPass Electronic Hall Pass System

<i>Source</i>	SmartPass Official Website "JTASD Approves Electronic Hall Pass System." Times News Online
<i>Image</i>	 https://raptortech.com/protect-your-school/smartpass-digital-hall-pass-solution/
<i>Details</i>	SmartPass is a digital hall pass management platform designed to replace paper hall passes with electronic tracking systems. Teachers can issue passes digitally, monitor active passes, and limit the number of students allowed in certain areas simultaneously. Schools using SmartPass have reported improved hallway management and reduced congestion (Hedes, 2026). The platform provides better organization and data collection compared to traditional systems because passes are electronically stored and monitored.

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<i>Analysis</i>	Although SmartPass improves organization and accountability, it still relies heavily on students honestly ending passes and correctly reporting their movement. In many cases, there is no physical verification that a student actually arrived at the destination. The system also requires school-wide internet access and student devices, increasing implementation costs. Additionally, subscription-based systems may not be affordable for all schools. These weaknesses support the need for a lower-cost RFID verification system that physically confirms student arrival.
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Prior Design Attempt #3 - RFID Attendance Tracking Systems

<i>Source</i>	RFID Attendance System Research
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<i>Image</i>	 https://www.rfidjournal.com/news/school-bus-solution-provides-app-based-visibility-for-parents/69812/
<i>Details</i>	RFID attendance systems use RFID cards and scanners to automatically record student attendance when students enter classrooms or buildings. Each student receives a unique RFID tag linked to a digital database. When scanned, the system records attendance information without requiring manual input from teachers. RFID technology is widely used because it improves efficiency and reduces attendance errors.
<i>Analysis</i>	RFID attendance systems effectively identify students, but they are generally designed only for classroom attendance rather than hall pass tracking. Most systems do not monitor destinations, time spent outside class, or return verification. While RFID identification is useful, previous attendance-focused systems fail to fully

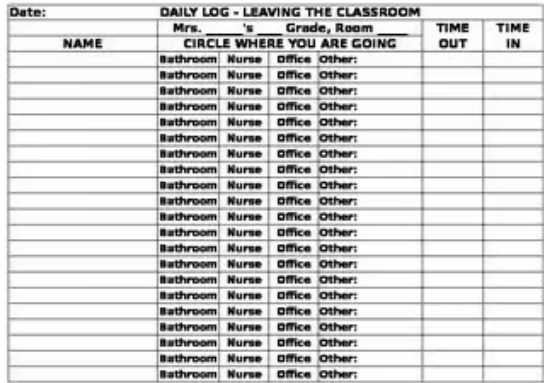
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	address the problem of student movement accountability throughout a school building. This project improves on those systems by combining RFID verification with time monitoring and destination tracking.
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Prior Design Attempt #4 - Manual Sign-Out Logs

Source	Education Week School Monitoring
Image	 https://www.teacherspayteachers.com/Product/Sign-Out-Log-for-Students-Leaving-the-Classroom-editable-2149167
Details	Manual sign-out logs require students to write their names, destinations, and departure times before leaving class. Teachers can later review these records to determine how long students were gone or



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	where they claimed to go. These systems are commonly used because they are inexpensive and easy to create without additional technology.
<i>Analysis</i>	Manual sign-out logs improve documentation slightly compared to verbal permission systems, but they still depend heavily on honesty and teacher supervision. Students may forget to record times, falsify information, or fail to sign back in after returning. Teachers must manually review records, which increases workload and reduces efficiency. The system also lacks automatic timing, real-time alerts, and identity verification, making it insufficient for solving the larger accountability problem.

Prior Design Attempt #5 - GPS Student Tracking Applications

<i>Source</i>	Education World GPS Tracking Overview
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<i>Image</i>	 https://pictortelematics.com/products/asset-tracking/id-card-with-gps-tracker
<i>Details</i>	GPS-based student tracking systems use mobile devices or school-issued technology to monitor student locations in real time. These systems are often used during transportation, field trips, or emergency situations because they provide detailed geographic location information. GPS technology allows administrators to monitor student movement remotely across larger areas.
<i>Analysis</i>	Although GPS systems provide accurate outdoor location tracking, they are not ideal for indoor school hallways where signals may become unreliable. GPS tracking also raises significant privacy concerns because students may be

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	continuously monitored throughout the day. Requiring smartphones or GPS-enabled devices increases costs and complexity for schools. Compared to GPS systems, RFID technology provides a simpler, lower-cost, and more privacy-focused solution for tracking student movement within school buildings.
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